

UNIT-2 Knowledge Representation

GTU Syllabus:

Knowledge Representation:

- ☐ Concept of Knowledge representation
- ☐ Introduction to Natural Language processing
- ☐ Concept of Pattern recognition
- ☐ Introduction to Expert systems

What is knowledge representation?

Humans are best at understanding, reasoning, and interpreting knowledge. Human knows things, which is knowledge and as per their knowledge they perform various actions in the real world. **But how machines do all these things comes under knowledge representation and reasoning.** Hence we can describe Knowledge representation as following:

- Knowledge representation and reasoning (KR, KRR) is the part of Artificial intelligence which concerned with AI agents thinking and how thinking contributes to intelligent behavior of agents.
- It is responsible for representing information about the real world so that a computer can understand and can utilize this knowledge to solve the complex real world problems such as diagnosis a medical condition or communicating with humans in natural language.
- It is also a way which describes how we can represent knowledge in artificial intelligence. Knowledge representation is not just storing data into some database, but it also enables an intelligent machine to learn from that knowledge and experiences so that it can behave intelligently like a human.

What to Represent:

Following are the kind of knowledge which needs to be represented in AI systems:

- **Object:** All the facts about objects in our world domain. E.g., Guitars contains strings, trumpets are brass instruments.
- **Events:** Events are the actions which occur in our world.
- **Performance:** It describe behavior which involves knowledge about how to do things.
- **Meta-knowledge:** It is knowledge about what we know.
- **Facts:** Facts are the truths about the real world and what we represent.
- **Knowledge-Base:** The central component of the knowledge-based agents is the knowledge base. It is represented as KB. The Knowledgebase is a group of the Sentences (Here, sentences are used as a technical term and not identical with the English language).

Knowledge: Knowledge is awareness or familiarity gained by experiences of facts, data, and situations. Following are the types of knowledge in artificial intelligence:

Types of knowledge

There are several ways to categorize knowledge in Artificial Intelligence. Here are some key types with examples:

1. Declarative Knowledge:

- **Definition:** This type of knowledge represents facts and statements about the world. It describes "what" is known, without detailing how to use it. It's often static and can be expressed in a declarative form.
- **Example:**
 - "Paris is the capital of France."
 - "Water boils at 100 degrees Celsius at standard atmospheric pressure."
 - In a medical AI system, knowing that "fever is a symptom of the flu."

2. Procedural Knowledge:

- **Definition:** This knowledge represents the steps or sequences of actions needed to perform a task or solve a problem. It focuses on "how" to do something. It's dynamic and action-oriented.
- **Example:**
 - Knowing the steps to change a flat tire on a car.
 - The algorithm a robot uses to assemble a product on an assembly line.
 - In natural language processing, the steps to parse a sentence grammatically.

3. Semantic Knowledge:

- **Definition:** This type of knowledge represents the relationships between concepts, entities, and words. It focuses on the meaning and understanding of information by organizing knowledge into structures like networks or hierarchies.
- **Example:**
 - Understanding that a "dog" is a type of "animal," and a "poodle" is a type of "dog."
 - In a semantic network for a medical AI, representing that "headache" is a "symptom" which can be "caused by" "stress" or "flu."
 - Using word embeddings in natural language processing where words with similar meanings have similar vector representations.

4. Structural Knowledge:

- **Definition:** This knowledge represents the organization and relationships within a problem or a system. It helps in decomposing complex problems into simpler parts.
- **Example:**
 - Understanding the hierarchical structure of a company's departments.
 - In biology, the taxonomic classification of living organisms (kingdom, phylum, class, etc.).
 - Knowing the components of a computer and how they are interconnected.

5. Meta-Knowledge:

- **Definition:** This is "knowledge about knowledge." It includes understanding what we know, what we don't know, and knowing which knowledge is relevant or important in a given situation.
- **Example:**
 - A medical diagnosis system knowing that certain symptoms are more critical than others for diagnosing a specific condition.
 - An AI planning system knowing which problem-solving strategies are likely to be most effective for different types of tasks.
 - Using strategies like repetition or visualization to remember new information.

6. Heuristic Knowledge:

- **Definition:** This is knowledge based on experience, rules of thumb, or best practices, which may not always be guaranteed to be optimal or correct but are often effective in practice.
- **Example:**
 - In game playing (like chess), using rules of thumb to evaluate board positions.
 - An expert system for troubleshooting computer problems might use rules like "if the computer doesn't turn on, first check the power cord."
 - In search algorithms, using heuristics to guide the search towards promising states.

7. Domain Knowledge:

- **Definition:** This is specific knowledge about a particular subject or area of expertise.
- **Example:**
 - A medical expert system having detailed knowledge about diseases, symptoms, and treatments.
 - A financial trading AI having knowledge about market trends, economic indicators, and trading strategies.

8. Common Sense Knowledge:

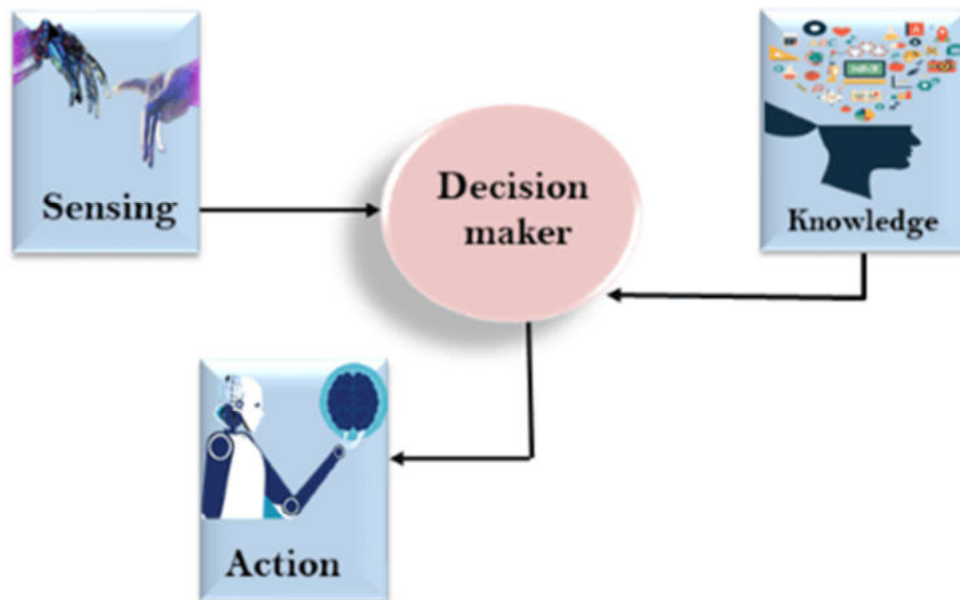
- **Definition:** This is the basic knowledge about the world that humans generally possess, enabling them to make reasonable inferences and understand everyday situations.
- **Example:**
 - Knowing that if you drop an object, it will fall to the ground.
 - Understanding that people generally feel happy when they receive a gift.
 - The fact that you cannot be in two places at once.

The relation between knowledge and intelligence:

Knowledge of real-worlds plays a vital role in intelligence and same for creating artificial intelligence. Knowledge plays an important role in demonstrating intelligent behavior in AI agents. An agent is only able to accurately act on some input when he has some knowledge or experience about that input.

Let's suppose if you met some person who is speaking in a language which you don't know, then how you will be able to act on that. The same thing applies to the intelligent behavior of the agents.

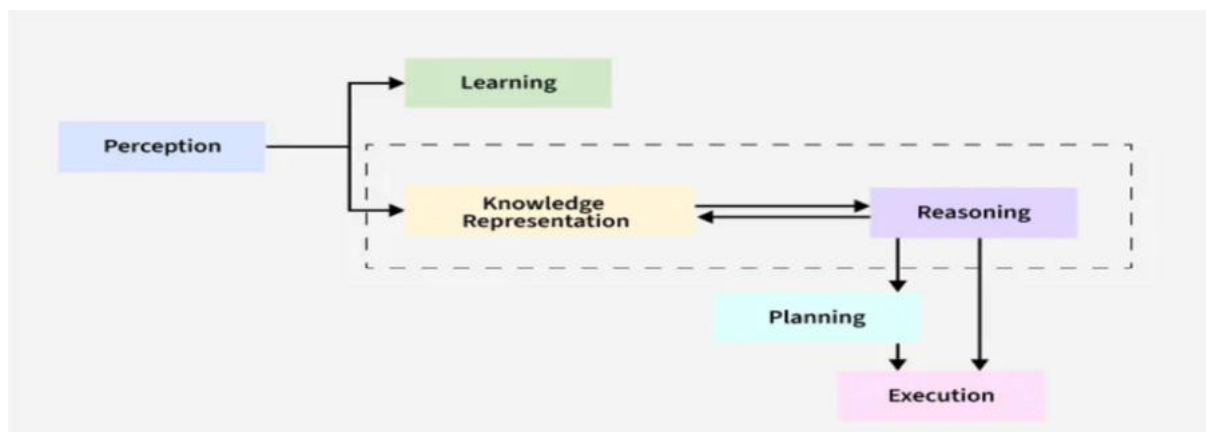
As we can see in below diagram, there is one decision maker which acts by sensing the environment and using knowledge. But if the knowledge part will not be present then, it cannot display intelligent behaviour.



AI knowledge cycle:

An Artificial intelligence system has the following components for displaying intelligent behaviour:

- Perception
- Learning
- Knowledge Representation and Reasoning
- Planning
- Execution



The above diagram is showing how an AI system can interact with the real world and what components help it to show intelligence. AI system has Perception component by which it retrieves information from its environment. It can be visual, audio or another form of sensory input. The learning component is responsible for learning from data captured by Perception component. In the complete cycle, the main components are knowledge representation and Reasoning. These two components are involved in showing the intelligence in machine-like humans. These two components are independent with each other but also coupled together. The planning and execution depend on analysis of Knowledge representation and reasoning.

1. Simple relational knowledge:

- It is the simplest way of storing facts which uses the relational method, and each fact about a set of the object is set out systematically in columns.
- This approach of knowledge representation is famous in database systems where the relationship between different entities is represented.
- This approach has little opportunity for inference.

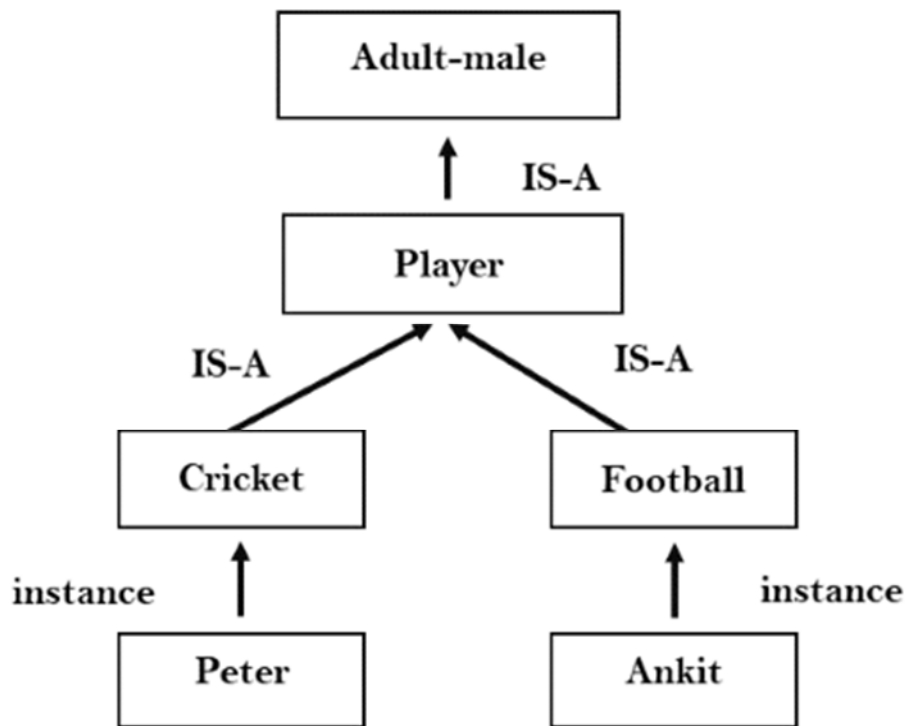
Example: The following is the simple relational knowledge representation.

Player	Weight	Age
Player1	65	23
Player2	58	18
Player3	75	24

2. Inheritable knowledge:

- In the inheritable knowledge approach, all data must be stored into a hierarchy of classes.
- All classes should be arranged in a generalized form or a hierarchal manner.
- In this approach, we apply inheritance property.
- Elements inherit values from other members of a class.
- This approach contains inheritable knowledge which shows a relation between instance and class, and it is called instance relation.
- Every individual frame can represent the collection of attributes and its value.
- In this approach, objects and values are represented in Boxed nodes.
- We use Arrows which point from objects to their values.

- **Example:**



3. Inferential knowledge:

- Inferential knowledge approach represents knowledge in the form of formal logics.
- This approach can be used to derive more facts.
- It guaranteed correctness.
- **Example:** Let's suppose there are two statements:

- Marcus is a man
 - All men are mortal
- Then it can represent as;

man(Marcus)

$\forall x = \text{man}(x) \text{ -----} \rightarrow \text{mortal}(x)$

4. Procedural knowledge:

- Procedural knowledge approach uses small programs and codes which describes how to do specific things, and how to proceed.
- In this approach, one important rule is used which is **If-Then rule**.

- In this knowledge, we can use various coding languages such as **LISP language** and **Prolog language**.
- We can easily represent heuristic or domain-specific knowledge using this approach.
- But it is not necessary that we can represent all cases in this approach.

Knowledge Representation system properties:

A good knowledge representation system must possess the following properties.

1. Representational Accuracy:

KR system should have the ability to represent all kind of required knowledge.

2. Inferential Adequacy:

KR system should have ability to manipulate the representational structures to produce new knowledge corresponding to existing structure.

3. Inferential Efficiency:

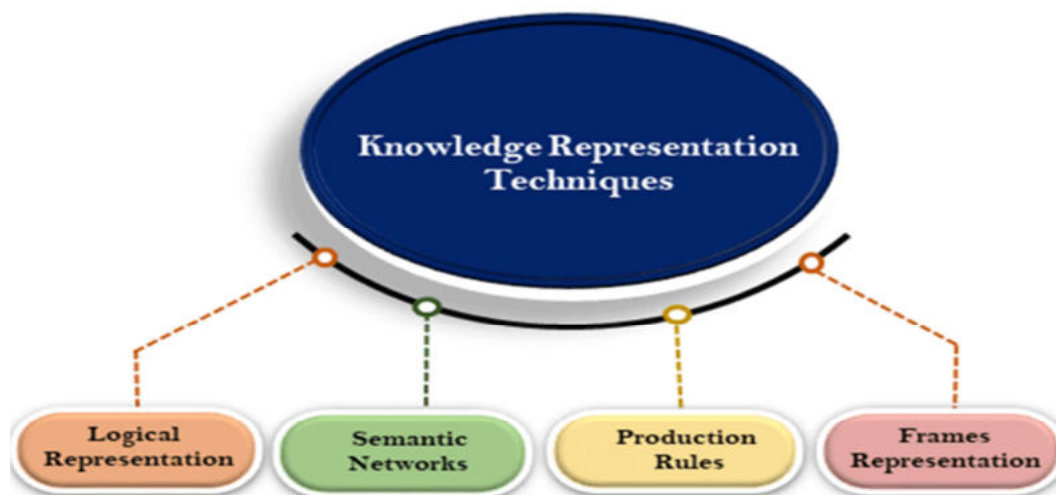
The ability to direct the inferential knowledge mechanism into the most productive directions by storing appropriate guides.

4. Acquisitional efficiency- The ability to acquire the new knowledge easily using automatic methods.

Techniques of knowledge representation:

There are mainly four ways of knowledge representation which are given as follows:

1. Logical Representation
2. Semantic Network Representation
3. Frame Representation
4. Production Rules



1. Logical Representation

Logical representation is a language with some concrete rules which deals with propositions and has no ambiguity in representation. Logical representation means drawing a conclusion based on various conditions. This representation lays down some important communication rules. It consists of precisely defined syntax and semantics which supports the sound inference. Each sentence can be translated into logics using syntax and semantics.

Syntax:

- Syntaxes are the rules which decide how we can construct legal sentences in the logic.
- It determines which symbol we can use in knowledge representation.
- How to write those symbols.

Semantics:

- Semantics are the rules by which we can interpret the sentence in the logic.
- Semantic also involves assigning a meaning to each sentence.

Logical representation can be categorised into mainly two logics:

- a. Propositional Logics
- b. Predicate logics

Advantages of logical representation:

1. Logical representation enables us to do logical reasoning.
2. Logical representation is the basis for the programming languages.

Disadvantages of logical Representation:

1. Logical representations have some restrictions and are challenging to work with.
2. Logical representation technique may not be very natural, and inference may not be so efficient.

Note: Do not be confused with logical representation and logical reasoning as logical representation is a representation language and reasoning is a process of thinking logically.

2. Semantic Network Representation

Semantic networks are alternative of predicate logic for knowledge representation. In Semantic networks, we can represent our knowledge in the form of graphical networks. This network consists of nodes representing objects and arcs which describe the relationship between those objects. Semantic networks can categorize the object in different forms and

can also link those objects. Semantic networks are easy to understand and can be easily extended.

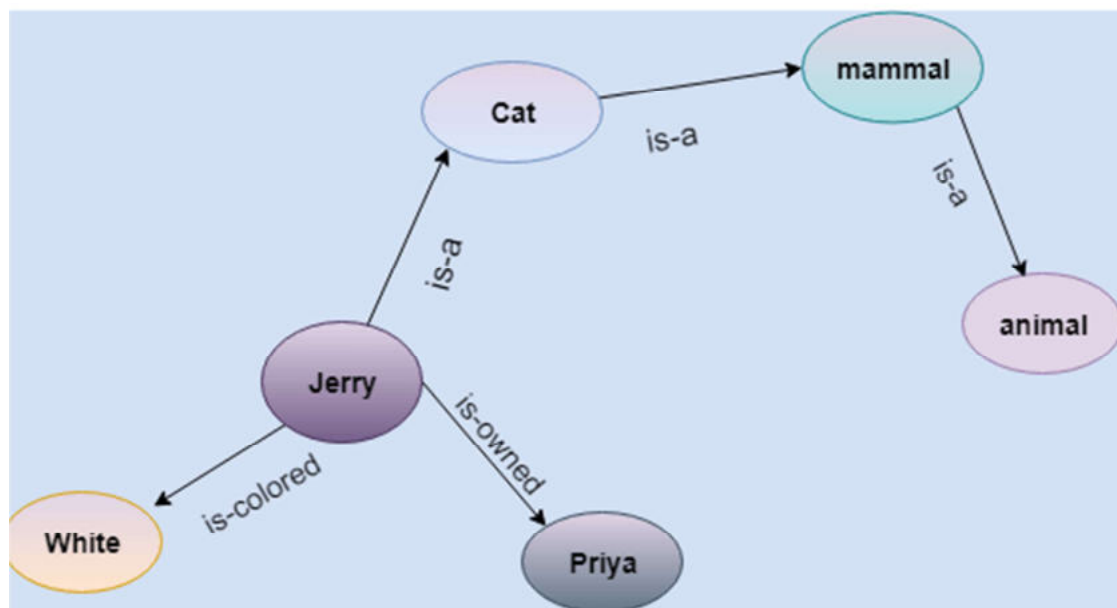
This representation consist of mainly two types of relations:

- a. IS-A relation (Inheritance)
- b. Kind-of-relation

Example: Following are some statements which we need to represent in the form of nodes and arcs.

Statements:

- a. Jerry is a cat.
- b. Jerry is a mammal
- c. Jerry is owned by Priya.
- d. Jerry is brown colored.
- e. All Mammals are animal.



In the above diagram, we have represented the different type of knowledge in the form of nodes and arcs. Each object is connected with another object by some relation.

Drawbacks in Semantic representation:

1. Semantic networks take more computational time at runtime as we need to traverse the complete network tree to answer some questions. It might be possible in the worst case scenario that after traversing the entire tree, we find that the solution does not exist in this network.

2. Semantic networks try to model human-like memory (Which has 1015 neurons and links) to store the information, but in practice, it is not possible to build such a vast semantic network.
3. These types of representations are inadequate as they do not have any equivalent quantifier, e.g., for all, for some, none, etc.
4. Semantic networks do not have any standard definition for the link names.
5. These networks are not intelligent and depend on the creator of the system.

Advantages of Semantic network:

1. Semantic networks are a natural representation of knowledge.
2. Semantic networks convey meaning in a transparent manner.
3. These networks are simple and easily understandable.

3. Frame Representation

A frame is a record like structure which consists of a collection of attributes and its values to describe an entity in the world. Frames are the AI data structure which divides knowledge into substructures by representing stereotypes situations. It consists of a collection of slots and slot values. These slots may be of any type and sizes. Slots have names and values which are called facets.

Facets: The various aspects of a slot is known as **Facets**. Facets are features of frames which enable us to put constraints on the frames. Example: IF-NEEDED facts are called when data of any particular slot is needed. A frame may consist of any number of slots, and a slot may include any number of facets and facets may have any number of values. A frame is also known as **slot-filter knowledge representation** in artificial intelligence.

Frames are derived from semantic networks and later evolved into our modern-day classes and objects. A single frame is not much useful. Frames system consist of a collection of frames which are connected. In the frame, knowledge about an object or event can be stored together in the knowledge base. The frame is a type of technology which is widely used in various applications including Natural language processing and machine visions.

Example: 1

Let's take an example of a frame for a book

Slots	Filters
Title	Artificial Intelligence
Genre	Computer Science

Author	Peter Norvig
Edition	Third Edition
Year	1996
Page	1152

Example 2:

Let's suppose we are taking an entity, Peter. Peter is an engineer as a profession, and his age is 25, he lives in city London, and the country is England. So following is the frame representation for this:

Slots	Filter
Name	Peter
Profession	Doctor
Age	25
Marital status	Single
Weight	78

Advantages of frame representation:

1. The frame knowledge representation makes the programming easier by grouping the related data.
2. The frame representation is comparably flexible and used by many applications in AI.
3. It is very easy to add slots for new attribute and relations.
4. It is easy to include default data and to search for missing values.
5. Frame representation is easy to understand and visualize.

Disadvantages of frame representation:

1. In frame system inference mechanism is not be easily processed.
2. Inference mechanism cannot be smoothly proceeded by frame representation.
3. Frame representation has a much generalized approach.

4. Production Rules

Production rules system consist of (**condition, action**) pairs which mean, "If condition then action". It has mainly three parts:

- The set of production rules
- Working Memory
- The recognize-act-cycle

In production rules agent checks for the condition and if the condition exists then production rule fires and corresponding action is carried out. The condition part of the rule determines which rule may be applied to a problem. And the action part carries out the associated problem-solving steps. This complete process is called a recognize-act cycle.

The working memory contains the description of the current state of problems-solving and rule can write knowledge to the working memory. This knowledge match and may fire other rules.

If there is a new situation (state) generates, then multiple production rules will be fired together, this is called conflict set. In this situation, the agent needs to select a rule from these sets, and it is called a conflict resolution.

Example:

- IF (at bus stop AND bus arrives) THEN action (get into the bus)
- IF (on the bus AND paid AND empty seat) THEN action (sit down).
- IF (on bus AND unpaid) THEN action (pay charges).
- IF (bus arrives at destination) THEN action (get down from the bus).

Advantages of Production rule:

1. The production rules are expressed in natural language.
2. The production rules are highly modular, so we can easily remove, add or modify an individual rule.

Disadvantages of Production rule:

1. Production rule system does not exhibit any learning capabilities, as it does not store the result of the problem for the future uses.
2. During the execution of the program, many rules may be active hence rule-based production systems are inefficient.

Propositional logic in Artificial intelligence

Propositional logic (PL) is the simplest form of logic where all the statements are made by propositions. A proposition is a declarative statement which is either true or false. It is a technique of knowledge representation in logical and mathematical form.

- a) It is Sunday.

b) The Sun rises from West (False proposition)

c) $3+3=7$ (False proposition)

d) 5 is a prime number.

Following are some basic facts about propositional logic:

- Propositional logic is also called Boolean logic as it works on 0 and 1.
- In propositional logic, we use symbolic variables to represent the logic, and we can use any symbol for a representing a proposition, such A, B, C, P, Q, R, etc.
- Propositions can be either true or false, but it cannot be both.
- Propositional logic consists of an object, relations or function, and **logical connectives**.
- These connectives are also called logical operators.
- The propositions and connectives are the basic elements of the propositional logic.
- Connectives can be said as a logical operator which connects two sentences.
- A proposition formula which is always true is called **tautology**, and it is also called a valid sentence.
- A proposition formula which is always false is called **Contradiction**.
- A proposition formula which has both true and false values is called
- Statements which are questions, commands, or opinions are not propositions such as "Where is Rohini", "How are you", "What is your name", are not propositions.

Syntax of propositional logic:

The syntax of propositional logic defines the allowable sentences for the knowledge representation. There are two types of Propositions:

- a. **Atomic Propositions**
- b. **Compound propositions**

- **Atomic Proposition:** Atomic propositions are the simple propositions. It consists of a single proposition symbol. These are the sentences which must be either true or false.

Example:

1. a) $2+2$ is 4, it is an atomic proposition as it is a **true** fact.
2. b) "The Sun is cold" is also a proposition as it is a **false** fact.

- **Compound proposition:** Compound propositions are constructed by combining simpler or atomic propositions, using parenthesis and logical connectives.

Example:

1. a) "It is raining today, and street is wet."
2. b) "Ankit is a doctor, and his clinic is in Mumbai."

Logical Connectives:

Logical connectives are used to connect two simpler propositions or representing a sentence logically. We can create compound propositions with the help of logical connectives. There are mainly five connectives, which are given as follows:

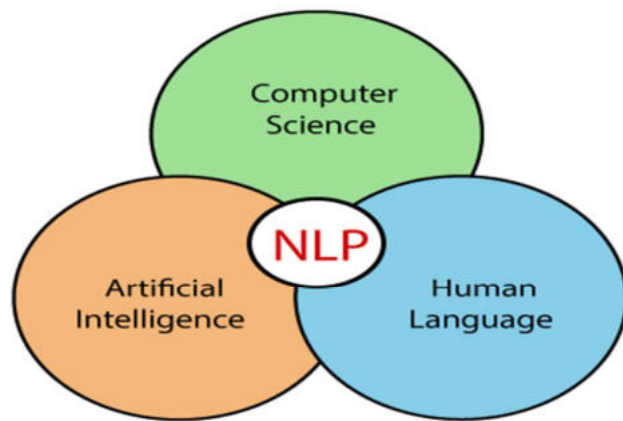
1. **Negation:** A sentence such as $\neg P$ is called negation of P. A literal can be either Positive literal or negative literal.
2. **Conjunction:** A sentence which has $\wedge$ connective such as, $P \wedge Q$ is called a conjunction.
Example: Rohan is intelligent and hardworking. It can be written as,
P= Rohan is intelligent,
Q= Rohan is hardworking. $\rightarrow P \wedge Q$.
3. **Disjunction:** A sentence which has $\vee$ connective, such as $P \vee Q$. is called disjunction, where P and Q are the propositions.
Example: "Ritika is a doctor or Engineer",
Here P= Ritika is Doctor. Q= Ritika is Doctor, so we can write it as $P \vee Q$.
4. **Implication:** A sentence such as $P \rightarrow Q$, is called an implication. Implications are also known as if-then rules. It can be represented as
 If it is raining, then the street is wet.
 Let P= It is raining, and Q= Street is wet, so it is represented as $P \rightarrow Q$
5. **Biconditional:** A sentence such as $P \leftrightarrow Q$ is a **Biconditional sentence, example If I am breathing, then I am alive**
Limitations of Propositional logic:
 - We cannot represent relations like ALL, some, or none with propositional logic.
Example:
 - a. **All the girls are intelligent.**
 - b. **Some apples are sweet.**
 - Propositional logic has limited expressive power.
 - In propositional logic, we cannot describe statements in terms of their properties or logical relationships.

Natural Language Processing:

What is NLP?

Definition: Natural Language Processing (NLP) is a branch of artificial intelligence (AI) that allows computers to understand, interpret, and generate human language. It's a field that bridges the gap between human communication and computer understanding, enabling machines to process and analyze text and speech.

NLP stands for Natural Language Processing, which is a part of Computer Science, Human language, and Artificial Intelligence. It is the technology that is used by machines to understand, analyse, manipulate, and interpret human's languages. It helps developers to organize knowledge for performing tasks such as translation, automatic summarization, Named Entity Recognition (NER), speech recognition, relationship extraction, and topic segmentation.



History of NLP

In the beginning of the year 1990s, NLP started growing faster and achieved good process accuracy, especially in English Grammar. In 1990 also, an electronic text introduced, which provided a good resource for training and examining natural language programs. Other factors may include the availability of computers with fast CPUs and more memory. The major factor behind the advancement of natural language processing was the Internet.

Now, modern NLP consists of various applications, like **speech recognition**, **machine translation**, and **machine text reading**. When we combine all these applications then it allows the artificial intelligence to gain knowledge of the world. Let's consider the example of AMAZON ALEXA, using this robot you can ask the question to Alexa, and it will reply to you.

Advantages of NLP

- NLP helps users to ask questions about any subject and get a direct response within seconds.
- NLP offers exact answers to the question means it does not offer unnecessary and unwanted information.
- NLP helps computers to communicate with humans in their languages.
- It is very time efficient.
- Most of the companies use NLP to improve the efficiency of documentation processes, accuracy of documentation, and identify the information from large databases.

Disadvantages of NLP

A list of disadvantages of NLP is given below:

- NLP may not show context.

- NLP is unpredictable
 - NLP may require more keystrokes.
 - NLP is unable to adapt to the new domain, and it has a limited function that's why NLP is built for a single and specific task only.
-

Components of NLP

There are the following two components of NLP -

1. Natural Language Understanding (NLU)

Natural Language Understanding (NLU) helps the machine to understand and analyse human language by extracting the metadata from content such as concepts, entities, keywords, emotion, relations, and semantic roles.

NLU mainly used in Business applications to understand the customer's problem in both spoken and written language.

NLU involves the following tasks -

- It is used to map the given input into useful representation.
- It is used to analyze different aspects of the language.

2. Natural Language Generation (NLG)

Natural Language Generation (NLG) acts as a translator that converts the computerized data into natural language representation. It mainly involves Text planning, Sentence planning, and Text Realization.

Note: The NLU is difficult than NLG.

Difference between NLU and NLG

NLU	NLG
NLU is the process of reading and interpreting language.	NLG is the process of writing or generating language.
It produces non-linguistic outputs from natural language inputs.	It produces constructing natural language outputs from non-linguistic inputs.

Applications of NLP

There are the following applications of NLP -

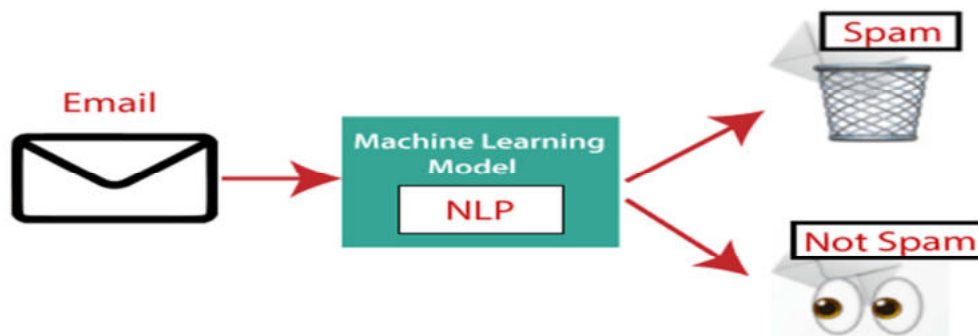
1. Question Answering

Question Answering focuses on building systems that automatically answer the questions asked by humans in a natural language.



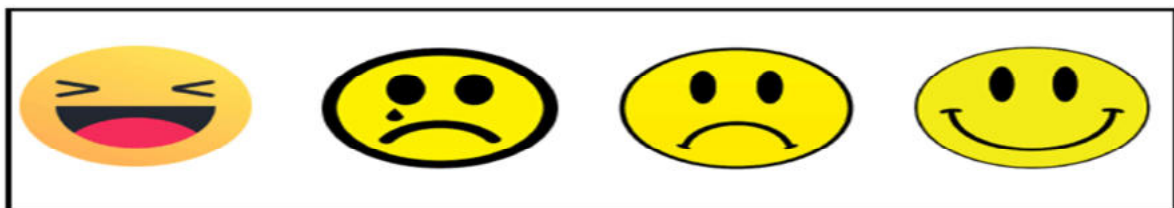
2. Spam Detection

Spam detection is used to detect unwanted e-mails getting to a user's inbox.



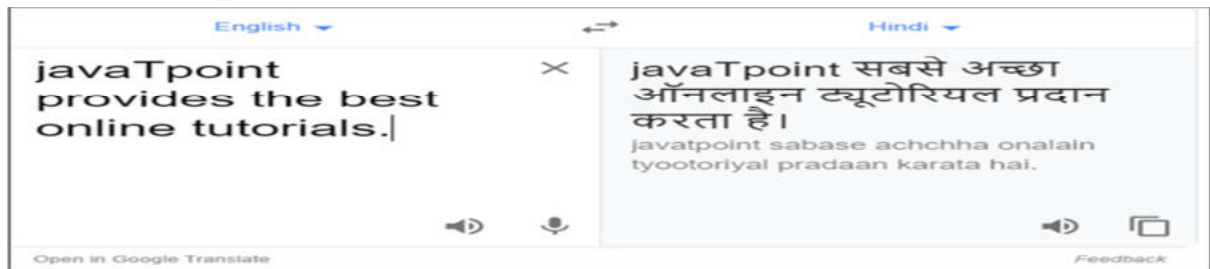
3. Sentiment Analysis

Sentiment Analysis is also known as **opinion mining**. It is used on the web to analyse the attitude, behaviour, and emotional state of the sender. This application is implemented through a combination of NLP (Natural Language Processing) and statistics by assigning the values to the text (positive, negative, or natural), identify the mood of the context (happy, sad, angry, etc.)



4. Machine Translation

Machine translation is used to translate text or speech from one natural language to another natural language.



Example: Google Translator

5. Spelling correction

Microsoft Corporation provides word processor software like MS-word, PowerPoint for the spelling correction.

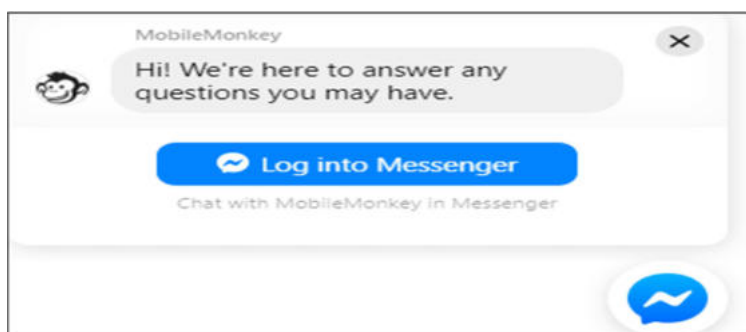


6. Speech Recognition

Speech recognition is used for converting spoken words into text. It is used in applications, such as mobile, home automation, video recovery, dictating to Microsoft Word, voice biometrics, voice user interface, and so on.

7. Chatbot

Implementing the Chatbot is one of the important applications of NLP. It is used by many companies to provide the customer's chat services.



8. Information extraction

Information extraction is one of the most important applications of NLP. It is used for extracting structured information from unstructured or semi-structured machine-readable documents.

9. Natural Language Understanding (NLU)

It converts a large set of text into more formal representations such as first-order logic structures that are easier for the computer programs to manipulate notations of the natural language processing.

How to build an NLP pipeline

There are the following steps to build an NLP pipeline -

Step1: Sentence Segmentation

Sentence Segment is the first step for building the NLP pipeline. It breaks the paragraph into separate sentences.

Example: Consider the following paragraph -

Independence Day is one of the important festivals for every Indian citizen. It is celebrated on the 15th of August each year ever since India got independence from the British rule. The day celebrates independence in the true sense.

Sentence Segment produces the following result:

1. "Independence Day is one of the important festivals for every Indian citizen."
2. "It is celebrated on the 15th of August each year ever since India got independence from the British rule."
3. "This day celebrates independence in the true sense."

Step2: Word Tokenization

Word Tokenizer is used to break the sentence into separate words or tokens.

Example:

Company offers Corporate Training, Summer Training, Online Training, and Winter Training.

Word Tokenizer generates the following result:

"company", "offers", "Corporate", "Training", "Summer", "Training", "Online", "Training", "and", "Winter", "Training", "."

Step3: Stemming

Stemming is used to normalize words into its base form or root form. For example, celebrates, celebrated and celebrating, all these words are originated with a single root word "celebrate." The big problem with stemming is that sometimes it produces the root word which may not have any meaning.

For Example, intelligence, intelligent, and intelligently, all these words are originated with a single root word "intelligen." In English, the word "intelligen" do not have any meaning.

Step 4: Lemmatization

Lemmatization is quite similar to the Stemming. It is used to group different inflected forms of the word, called Lemma. The main difference between Stemming and lemmatization is that it produces the root word, which has a meaning.

For example: In lemmatization, the words intelligence, intelligent, and intelligently has a root word intelligent, which has a meaning.

Step 5: Identifying Stop Words

In English, there are a lot of words that appear very frequently like "is", "and", "the", and "a". NLP pipelines will flag these words as stop words. **Stop words** might be filtered out before doing any statistical analysis.

Example: He is a good boy.

Note: When you are building a rock band search engine, then you do not ignore the word "The."

Step 6: Dependency Parsing

Dependency Parsing is used to find that how all the words in the sentence are related to each other.

Step 7: POS tags

POS stands for parts of speech, which includes Noun, verb, adverb, and Adjective. It indicates that how a word functions with its meaning as well as grammatically within the sentences. A word has one or more parts of speech based on the context in which it is used.

Example: "Google" something on the Internet.

In the above example, Google is used as a verb, although it is a proper noun.

Step 8: Named Entity Recognition (NER)

Named Entity Recognition (NER) is the process of detecting the named entity such as person name, movie name, organization name, or location.

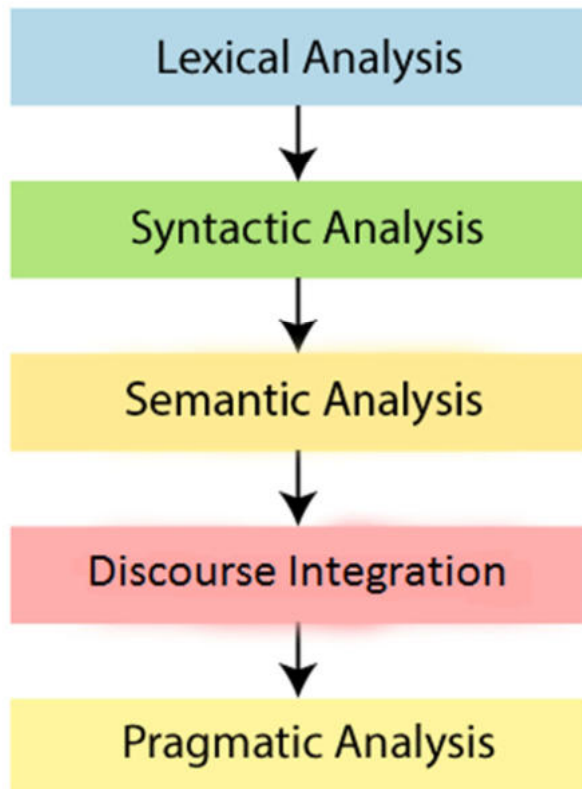
Example: Steve Jobs introduced iPhone at the Macworld Conference in San Francisco, California.

Step 9: Chunking

Chunking is used to collect the individual piece of information and grouping them into bigger pieces of sentences.

Phases of NLP (Imp question)

There are the following five phases of NLP:



1. Lexical Analysis and Morphological

The first phase of NLP is the Lexical Analysis. This phase scans the source code as a stream of characters and converts it into meaningful lexemes. It divides the whole text into paragraphs, sentences, and words.

2. Syntactic Analysis (Parsing)

Syntactic Analysis is used to check grammar, word arrangements, and shows the relationship among the words.

Example: Agra goes to the Poonam

In the real world, Agra goes to the Poonam, does not make any sense, so this sentence is rejected by the Syntactic analyzer.

3. Semantic Analysis

Semantic analysis is concerned with the meaning representation. It mainly focuses on the literal meaning of words, phrases, and sentences.

4. Discourse Integration

Discourse Integration depends upon the sentences that precedes it and also invokes the meaning of the sentences that follow it.

5. Pragmatic Analysis

Pragmatic is the fifth and last phase of NLP. It helps you to discover the intended effect by applying a set of rules that characterize cooperative dialogues.

For Example: "Open the door" is interpreted as a request instead of an order.

Why NLP is difficult? (Why NLP is Hard?)

NLP is difficult because Ambiguity and Uncertainty exist in the language.

Ambiguity

There are the following three ambiguity -

- **Lexical Ambiguity**

Lexical Ambiguity exists in the presence of two or more possible meanings of the sentence within a single word.

Example:

Manya is looking for a **match**.

In the above example, the word match refers to that either Manya is looking for a partner or Manya is looking for a match. (Cricket or other match)

- **Syntactic Ambiguity**

Syntactic Ambiguity exists in the presence of two or more possible meanings within the sentence.

Example:

I saw the girl with the binocular.

In the above example, did I have the binoculars? Or did the girl have the binoculars?

- **Referential Ambiguity**

Referential Ambiguity exists when you are referring to something using the pronoun.

Example: Kiran went to Sunita. She said, "I am hungry."

In the above sentence, you do not know that who is hungry, either Kiran or Sunita.

NLP Libraries for Programming

Scikit-learn: It provides a wide range of algorithms for building machine learning models in Python.

Natural language Toolkit (NLTK): NLTK is a complete toolkit for all NLP techniques.

Pattern: It is a web mining module for NLP and machine learning.

TextBlob: It provides an easy interface to learn basic NLP tasks like sentiment analysis, noun phrase extraction, or pos-tagging.

Quepy: Quepy is used to transform natural language questions into queries in a database query language.

SpaCy: SpaCy is an open-source NLP library which is used for Data Extraction, Data Analysis, Sentiment Analysis, and Text Summarization.

Gensim: Gensim works with large datasets and processes data streams.

Difference between Natural language and Computer Language

Natural Language	Computer Language
Natural language has a very large vocabulary.	Computer language has a very limited vocabulary.
Natural language is easily understood by humans.	Computer language is easily understood by the machines.
Natural language is ambiguous in nature.	Computer language is unambiguous.

What Is Pattern Recognition?

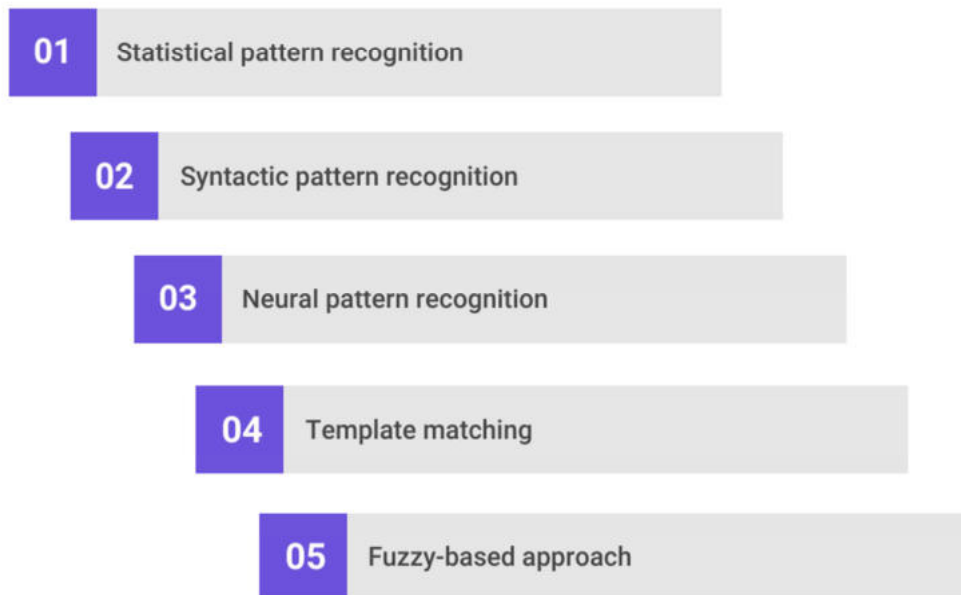
Pattern recognition is a data analysis process that uses machine learning algorithms to classify input data into objects, classes, or categories based on recognized patterns, features, or regularities in data. It has several applications in the fields of astronomy, medicine, robotics, and satellite remote sensing, among others.

Pattern recognition involves two primary classification methods:

- **Supervised classification:** In a supervised pattern recognition method, a human trains a computer algorithm to recognize patterns based on predefined labeled datasets. Once the pattern is identified, the method subsequently classifies new data.
- **Unsupervised classification:** In unsupervised classification, the model learns independently without any direct guidance from a human. The computer algorithm identifies correlations between multiple data elements (inputs) based on their similarity score and performs data classification.

Methods of pattern Recognition:

Methods of Pattern Recognition



1. Statistical pattern recognition

This pattern recognition approach uses historical statistical data that learns from patterns and examples. The method collects observations and processes them to define a model. This model then generalizes over the collected observations and applies the rules to new datasets or examples.

2. Syntactic pattern recognition

Syntactic pattern recognition involves complex patterns that can be identified using a hierarchical approach. Patterns are established based on the way primitives (e.g., letters in a word) interact with each other. An example of this could be how primitives are assembled in words and sentences. Such training samples will enable the development of grammatical rules that demonstrate how sentences will be read in the future.

3. Neural pattern recognition

This method **uses** artificial neural networks (ANN) and learns from complex and non-linear input/output relations, adapts to data, and detects patterns. The most popular and effective method in neural networks is the feed-forward method. In this method, learning happens by giving feedback to input patterns. This is much like humans learning from their past experiences and mistakes. The ANN-based model is rated as the most expensive pattern recognition method compared to other methods due to the computing resources involved in the process.

4. Template matching

Template matching is one of the simplest of all pattern recognition approaches.

Here, the similarity between two entities is determined by matching the sample with the reference template. Such methods are typically used in digital image processing, where small sections of an image are matched to a stored template image. Some of its real-world examples include medical image processing, face recognition, and robot navigation.

5. Fuzzy-based approach

In the fuzzy approach, a set of patterns are partitioned based on the similarity in the features of the patterns. When the unique features of a pattern are correctly detected, data can be easily classified into that known feature space. Even the human visual system sometimes fails to recognize certain components despite scanning objects for a long time. The same holds true for the digital world, where algorithms cannot figure out the exact nature of an object. Hence, the fuzzy approach aims to classify objects based on several similar features in the detected patterns.

6. Hybrid approach

A hybrid approach employs a combination of the above methods to take advantage of all these methods. It employs multiple classifiers to detect patterns

where each classifier is trained on a specific feature space. A conclusion is drawn based on the results accumulated from all the classifiers.

How Does Pattern Recognition Work?

Pattern recognition is applied for data of all types, including image, video, text, and audio. As the pattern recognition model can identify recurring patterns in data, predictions made by such models are quite reliable.

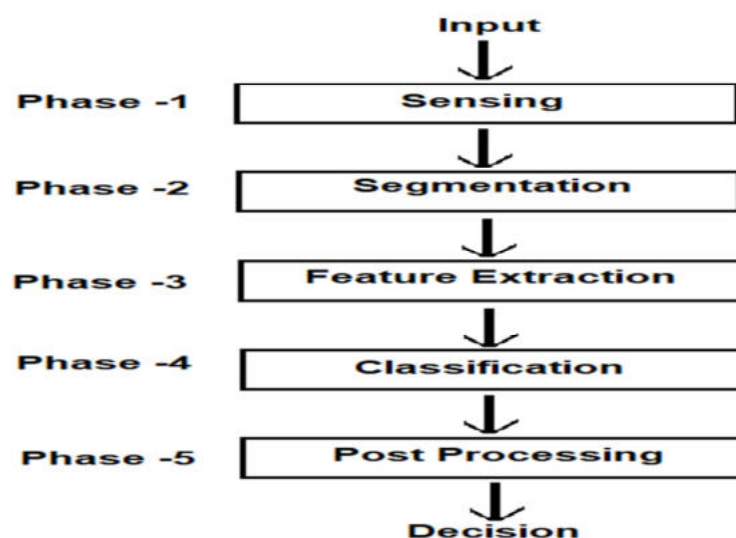
Pattern recognition involves three key steps: analyzing the input data, extracting patterns, and comparing it with the stored data. The process can be further divided into two phases:

1. **Explorative phase:** In this phase, computer algorithms tend to explore data patterns.
2. **Descriptive phase:** Here, algorithms group and attribute identified patterns to new data.

Approaches for Pattern Recognition Systems can be represented by distinct phases, as Pattern Recognition Systems can be divided into the following components.

Phases in Pattern Recognition System

- **Phase 1:** Convert images sounds or other inputs into signal data.
- **Phase 2:** Isolate the sensed objects from the background.
- **Phase 3:** Measure the object's properties that are useful for classification.
- **Phase 4:** Assign the sensed object to a category.
- **Phase 5:** Take other considerations to decide on appropriate action.



Phases in Pattern Recognition

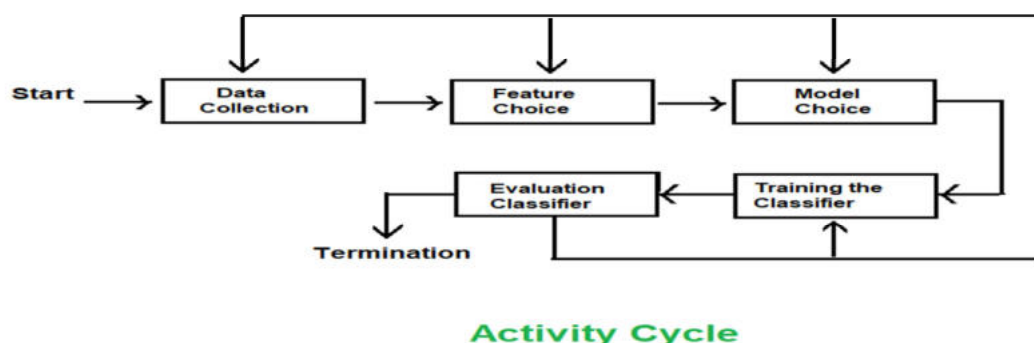
Problems Solved by these Phases

- **Sensing:** It deals with problem arises in the input such as its bandwidth, resolution, sensitivity, distortion, signal-to-noise ratio, latency, etc.
- **Segmentation and Grouping:** Deepest problems in pattern recognition that deals with the problem of recognizing or grouping together the various parts of an object.
- **Feature Extraction:** It deals with the characterization of an object so that it can be recognized easily by measurements. Those objects whose values are very similar for the objects are considered to be in the same category, while those whose values are quite different for the objects are placed in different categories.
- **Classification:** It deals with assigning the object to their particular categories by using the feature vector provided by the feature extractor and determining the values of all of the features for a particular input.
- **Post Processing:** It deals with action decision-making by using the output of the classifier. Action such as minimum-error-rate classification will minimize the total expected cost.

Activities for Designing the Pattern Recognition Systems

There are various sequences of activities that are used for designing the Pattern Recognition Systems. These activities are as follows:

- Data Collection
- Feature Choice
- Model Choice
- Training
- Evaluation



- **Data Collection:** This is the initial step where you gather the data that you'll use to train and test your pattern recognition system. The quality and quantity of data collected are crucial factors in the success of your system.
- **Feature Choice:** Features are the characteristics or attributes of the data that are relevant for pattern recognition. In this step, you decide which features or variables to use from your data. Feature selection is essential to reduce dimensionality and focus on the most informative aspects of your data.
- **Model Choice:** This step involves selecting the appropriate pattern recognition model or algorithm. The choice of model depends on the nature of the data and the problem at hand.
- **Training:** Once the model is chosen, it needs to be trained using a labeled dataset. During training, the model learns to identify patterns in the data and make predictions based on the features.
- **Evaluation:** After training, the performance of the pattern recognition system is assessed using a separate dataset that the model has not seen before. This evaluation dataset is used to measure the system's ability to correctly recognize patterns and make predictions.
- **There are typically four main phases** in the pattern recognition process: preprocessing, training, testing, and deployment. These phases involve a series of activities that are designed to develop and evaluate a pattern recognition system.
- **Preprocessing:** Preprocessing is the process of preparing the data for analysis. This may involve cleaning the data, scaling the data, or transforming the data in some way to make it more suitable for analysis.
- **Training:** Training is the process of fitting a model to the data. This typically involves selecting a model, choosing appropriate hyperparameters, and optimizing the model's parameters to minimize a loss function.
- **Testing:** Testing is the process of evaluating the performance of the model on a held-out dataset. This allows us to estimate the generalization performance of the model and to compare the performance of different models.
- **Deployment:** Deployment is the process of deploying the trained model in a production environment. This may involve integrating the model into an existing system or building a new system based on the model.

Applications of Pattern Recognition

Pattern recognition is a key technique in many fields of science, engineering, and everyday applications. Here's a list of common and important applications:

1. Computer Vision

- Facial recognition (e.g., unlocking phones)
 - Object detection (e.g., autonomous vehicles)
 - Optical character recognition (OCR) (e.g., scanning printed documents)
 - Medical imaging (e.g., tumor detection in X-rays or MRIs)
-

2. Speech and Audio Processing

- Voice recognition (e.g., Siri, Alexa)
 - Speaker identification
 - Speech-to-text transcription
 - Music genre classification
-

3. Natural Language Processing (NLP)

- Text classification (e.g., spam detection)
 - Sentiment analysis
 - Language translation
 - Chatbots and virtual assistants
-

4. Biometric Identification

- Fingerprint recognition
 - Iris and retina scanning
 - Gait recognition
-

5. Financial Applications

- Fraud detection
- Stock market pattern analysis
- Credit scoring

6. Robotics and Automation

- Gesture recognition
- Path planning using pattern analysis
- Sensor data interpretation

7. Cybersecurity

- Anomaly detection in network traffic
- Malware detection
- User behavior profiling

8. Industrial and Quality Control

- Defect detection in manufacturing
- Pattern-based predictive maintenance
- Automated inspection systems

9. Healthcare

- Disease prediction and diagnosis
- Genomic pattern analysis
- Patient monitoring systems

10. Environmental and Agricultural Monitoring

- Crop disease detection
- Weather pattern analysis
- Land use classification using satellite imagery

Introduction to Expert System:

Definition: An expert system is a computer program that uses artificial intelligence (AI) technologies to simulate the judgment and behaviour of a human or an organization that has expertise and experience in a particular field.

Expert systems are a crucial subset of artificial intelligence (AI) that simulates the decision-making ability of a human expert. These systems use a knowledge base filled with domain-specific information and rules to interpret and solve complex problems. For example, a medical expert system can analyse a patient's symptoms and suggest possible diagnoses or treatments. Similarly, a financial expert system can evaluate market trends and recommend investment strategies.

The key idea behind expert systems is to preserve and replicate human expertise. This is especially useful in fields where expert knowledge is scarce or expensive.

Why Are Expert Systems Important?

Expert systems are a game-changer in AI because they:

1. **Preserving Expertise:** They capture the knowledge of human experts and store it in a digital format. This ensures that valuable expertise isn't lost when an expert retires or leaves.
2. **Improving Decision-Making:** By relying on data and rules, expert systems provide consistent and unbiased recommendations.
3. **Saving Time and Money:** They automate tasks that would otherwise require human intervention, reducing costs and increasing efficiency.
4. **Accessibility:** Expert systems make expert-level knowledge available to non-experts, democratizing access to specialized information.

For instance, in the 1970s, the **MYCIN** system was developed to diagnose bacterial infections. While it was never used in real hospitals, it demonstrated how expert systems could assist doctors in making accurate diagnoses.

Components and Architecture of Expert System:

An expert system is made up of several interconnected components, each playing a crucial role in its functionality.

1. Knowledge Base: The Heart of the System

- The knowledge base is the heart of an expert system. It contains all the facts, rules, and expert knowledge related to a specific domain. Think of it as a library filled with textbooks, research papers, and expert opinions. *The **accuracy and completeness of the knowledge base directly impact the system's performance. If the knowledge is outdated or incomplete, the system's recommendations may be flawed.***
- *In a financial expert system, the knowledge base might include rules for detecting fraudulent transactions, such as "If a transaction exceeds \$10,000 and occurs in a foreign country, flag it for review."*

2. Inference Engine: The Brain Behind the Decisions

- The **inference engine** is the brain of the expert system. It processes the information stored in the knowledge base to draw conclusions or make recommendations. **The**

inference engine uses reasoning strategies (like forward chaining or backward chaining) to analyze data and apply rules.

- **Forward Chaining:** Starts with available data and works toward a conclusion. For example, *“If the temperature is high and the patient has a cough, diagnose a respiratory infection.”*
- **Backward Chaining:** Starts with a goal and works backward to find supporting evidence. For example, *“If the goal is to diagnose diabetes, check for symptoms like frequent urination and high blood sugar.”*

3. User Interface: Bridging the Gap Between System and User

The **user interface** is the bridge that allows users to interact with the expert system. It's designed to be intuitive and user-friendly, ensuring **that even non-experts can use the system effectively**. Users provide a **query** (problem or question), and the system processes the request. The system then delivers **advice** or recommendations back to the user.

4. Explanation Module: Building Trust Through Transparency

The **explanation module** is a critical feature that explains how the system arrived at a particular conclusion. It's like a teacher **showing their work when solving a math problem**. This module provides users with a clear, step-by-step explanation of the system's reasoning.

- This transparency is especially important in fields like healthcare and finance, where decisions can have significant consequences.
- **Example:** A medical expert system might explain, *“I diagnosed pneumonia because the patient has a fever, cough, and abnormal chest X-ray.”*

5. Knowledge Acquisition Module: Keeping the System Up-to-Date

The **knowledge acquisition module** is responsible for updating and expanding the knowledge base. It ensures that the system stays current with the latest information and trends. Without regular updates, the system's knowledge base can become outdated, reducing its effectiveness.

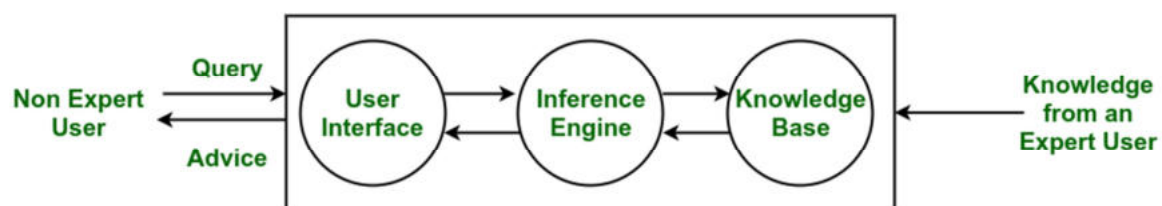


Fig. Expert Systems in AI : Architecture

The **working mechanism of an expert system** begins when a **non-expert user** submits a **query** through the **user interface**.

- This query is then processed by the **inference engine**, which applies **logical rules** and **reasoning techniques** to analyze the input.
- The **inference engine** interacts with the **knowledge base**, retrieving relevant **facts, rules, and heuristics** contributed by **expert users**.

- Based on this **structured knowledge**, the system derives **conclusions** and formulates an appropriate **response**.

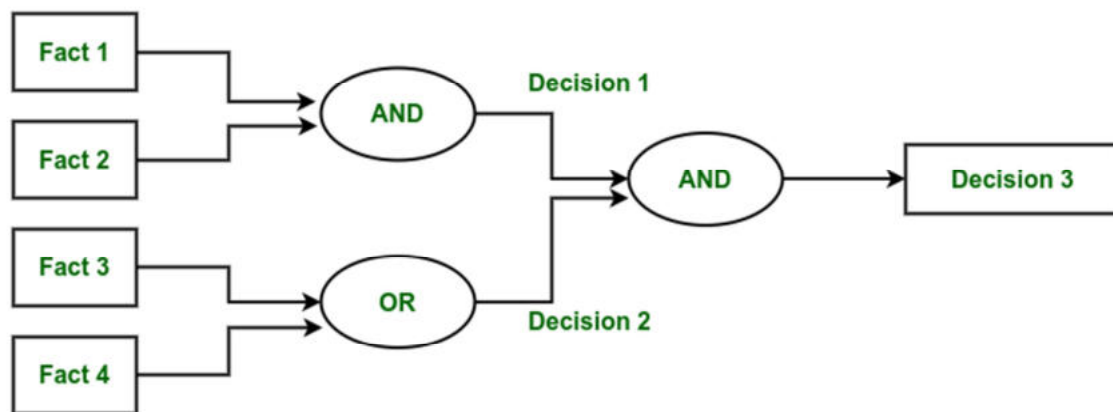
Finally, the **expert system** provides **advice** or **recommendations** to the user, assisting in **decision-making** or **problem-solving** without requiring direct **human expert** intervention.

Reasoning Strategies used by Inference Engine

Forward Chaining and **Backward Chaining**, which are two fundamental methods for processing information and solving problems in an expert system:

1. Forward Chaining

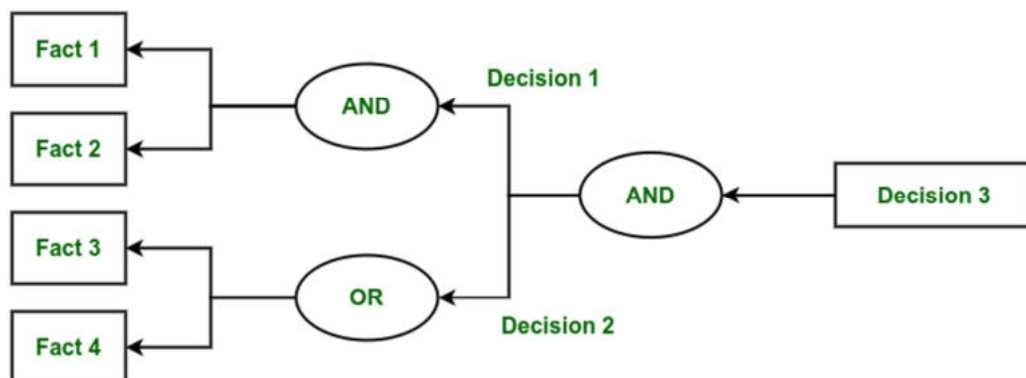
This is a **data-driven** reasoning approach where the system starts with the available facts and applies rules to infer new facts or conclusions. It's typically used to predict outcomes or determine what will happen next. An example given is predicting stock market movements.



Forward Chaining

2. Backward Chaining

This is a **goal-driven** reasoning approach where the system starts with a hypothesis or a goal (something to prove) and works backward to determine which facts or conditions would support that conclusion. It's often used to diagnose issues by determining the cause of an observed effect. The examples provided include diagnosing medical conditions like stomach pain, blood cancer, or dengue.



Backward Chaining

How These Components Work Together?

Imagine a medical expert system designed to diagnose diseases:

1. **Input:** A patient reports symptoms like fever, cough, and fatigue through the user interface.
2. **Processing:** The inference engine analyzes the symptoms using rules from the knowledge base.
3. **Output:** The system suggests a possible diagnosis, such as pneumonia.
4. **Explanation:** The explanation module provides a detailed explanation, such as “*The diagnosis is based on the presence of fever, cough, and abnormal chest X-ray results.*”
5. **Update:** The knowledge acquisition module adds new data, such as recent research on pneumonia treatments, to keep the system up-to-date.

Types of Expert Systems in AI

Depending on their structure and application, expert systems can be categorized into different types.

1. Rule-Based Expert Systems

One of the most common types is **Rule-Based Expert Systems**, which rely on **if-then rules** to process information and make decisions. These rules are typically crafted by domain experts and serve as the system’s reasoning mechanism. A well-known example is **MYCIN**, an early medical diagnosis system that identified bacterial infections.

2. Frame-Based Expert Systems

Another category is **Frame-Based Expert Systems**, which organize knowledge using **frames**, similar to objects in programming. These frames store attributes and values related to specific concepts, making them useful in natural language processing and other knowledge representation tasks.

3. Fuzzy Logic Systems

For situations involving **uncertainty and imprecision**, fuzzy logic Systems come into play. These systems don’t operate on strict true/false values but instead allow for degrees of truth. **Fuzzy control systems**, commonly used in household appliances like **washing machines and air conditioners**, leverage this approach to optimize performance based on variable input conditions.

4. Neural Network-Based Expert Systems

Integrate artificial neural networks to **learn patterns from data** and improve decision-making. These systems are widely used in applications like **image recognition and speech processing**, where traditional rule-based approaches might struggle.

5. Neuro-Fuzzy Expert Systems

A more advanced hybrid approach is **Neuro-Fuzzy Expert Systems**, which merge the learning capabilities of neural networks with the uncertainty-handling strengths of **fuzzy logic**. These systems are particularly useful in **financial forecasting** and **automated control systems**, where both structured learning and flexible reasoning are necessary.

Applications of Expert Systems

1. **Medical Diagnosis:** Expert systems assist doctors by analyzing symptoms and medical history to suggest possible diagnoses or treatment options. For example, MYCIN, an early expert system, helped identify bacterial infections and recommend antibiotics.
2. **Financial Services:** In finance, expert systems are used for credit scoring, fraud detection, and investment advice. They analyze financial data and patterns to make informed decisions.
3. **Technical Support:** Expert systems can troubleshoot and provide solutions for technical issues. They guide users through problem-solving steps based on pre-defined rules and knowledge.
4. **Manufacturing:** In manufacturing, expert systems help optimize production processes, perform quality control, and manage inventory by analyzing data and making recommendations.
5. **Education and Training:**
 - Intelligent Tutoring Systems:** These systems can adapt to a student's learning pace and provide personalized instruction and feedback.
 - Simulation and Training:** Expert systems can create realistic simulations for training in various fields, such as aviation or emergency response.
6. **Other Areas:**
 - Agriculture:** Expert systems can assist with crop management by analyzing soil conditions, weather data, and pest information.
 - Legal Decision-Making:** Some systems are used to analyze legal cases and provide insights or recommendations.
 - Environmental Monitoring:** Expert systems can analyze environmental data to detect pollution or predict ecological changes.
 - Resource Management:** They can help in optimizing the allocation of resources in various domains.

Benefits of Expert Systems (Advantages)

1. **Consistency:** Expert systems provide consistent and reliable recommendations, reducing the variability that can occur with human decision-making.
2. **Availability:** They are available 24/7 and can handle multiple queries simultaneously, providing timely assistance and support.
3. **Cost-Effectiveness:** By automating expert-level decision-making, organizations can save on the costs associated with hiring and training human experts.
4. **Knowledge Preservation:** Expert systems preserve valuable knowledge and expertise, making it accessible even if the original experts are no longer available.

Limitations of Expert Systems (Disadvantages)

1. **Knowledge Limitation:** The effectiveness of an expert system depends on the completeness and accuracy of the knowledge base. If the knowledge is outdated or incomplete, the system's performance may be compromised.
2. **Lack of Flexibility:** Expert systems are limited to the rules and knowledge they are programmed with. They may struggle with novel or ambiguous situations that fall outside their predefined rules.
3. **Maintenance:** Regular updates and maintenance are required to keep the knowledge base current and relevant, which can be resource-intensive.

Types of Language Models:

Language models can be categorized based on architecture, purpose, or training method. Here's an overview of the main **types of language models**:

1. Based on Architecture

a. Statistical Language Models (Traditional)

- **N-gram models**: Predict the next word based on the previous $n-1$ words.
- **Hidden Markov Models (HMMs)**: Use states and transitions for sequence prediction.

b. Neural Language Models

- **Feedforward Neural Networks**: Early neural models that use a fixed context window.
 - **Recurrent Neural Networks (RNNs)**: Capture sequential dependencies.
 - **LSTM (Long Short-Term Memory)** and **GRU (Gated Recurrent Unit)**: Handle long-term dependencies better than vanilla RNNs.
 - **Transformer-based Models**: State-of-the-art architecture for language tasks.
 - Examples: **BERT, GPT, T5, RoBERTa**
-

2. Based on Training Objective

a. Causal Language Models (Autoregressive)

- Predict the next word in a sequence.
- Example: **GPT** (Generative Pre-trained Transformer)

b. Masked Language Models

- Predict missing words in a sequence (masked inputs).
- Example: **BERT** (Bidirectional Encoder Representations from Transformers)

c. Seq2Seq (Encoder-Decoder) Models

- Map one sequence to another, common in translation and summarization.
 - Example: **T5, BART**
-

3. Based on Purpose

a. Generative Models

- Generate coherent text or code.
- Examples: **GPT, LLaMA, Claude**

b. Discriminative Models

- Classify or evaluate language input.
- Example: **RoBERTa, DistilBERT**

c. Multimodal Models

- Understand or generate across multiple data types (e.g., text + image).
- Examples: **GPT-4 (with vision), CLIP, Flamingo**

d. Instruction-tuned or Chat Models

- Trained to follow human instructions or engage in dialogue.
- Examples: **ChatGPT, Claude, Mistral-Instruct**

Question Bank:

1. What is knowledge representation? Explain types of knowledge.
2. What is the relation between knowledge and intelligence?
3. Explain AI knowledge cycle.
4. Explain Knowledge representation properties.
5. Describe techniques of knowledge representation.
6. What is Propositional logic in Artificial intelligence?
7. What is NLP? describe history of NLP.
8. Explain component of NLP.
9. How to built NLP pipeline.
10. Explain Phases of NLP.
11. Why NLP is difficult (hard)?
12. Difference between natural language and computer language
13. what is pattern recognition? Explain methods of pattern recognition.
14. How Does Pattern Recognition Work?
15. Explain phase of pattern recognition.
16. Write Applications of pattern recognition
17. What is Expert system (ES)? Describe Architecture and component of ES.
18. Explain Types of Expert system.
19. Write Advantages and disadvantages of Expert system

3 Marks Questions:

1. Briefly explain the significance of Knowledge Representation in Artificial Intelligence.
2. Differentiate between declarative and procedural knowledge representation.

3. What is the basic concept behind Natural Language Processing (NLP)? Give a simple example of an NLP task.
4. Define Pattern Recognition and provide a real-world application.
5. What is the core idea behind an Expert System? Mention one key component. [[Refer Pdf for Expert system](#)]

4 Marks Questions:

1. Discuss the challenges involved in representing knowledge effectively.
2. Explain the different levels of Natural Language Processing with a brief description of each.
3. Describe the general steps involved in a typical Pattern Recognition system.
4. Explain the architecture of a basic Expert System, highlighting the roles of the knowledge base and inference engine.
5. Compare and contrast rule-based and frame-based knowledge representation techniques.

7 Marks Questions:

1. Elaborate on the concept of Knowledge Representation, discussing its importance and various approaches (you can briefly mention a few like semantic networks, frames, rules).
2. Discuss the field of Natural Language Processing, covering its main goals, challenges, and some of its key applications.
3. Explain the process of Pattern Recognition in detail, including the different stages involved and the factors that influence its performance.
4. Describe Expert Systems in detail, including their components, advantages, limitations, and the process of building one. Provide a relevant example of an Expert System in a specific domain.
5. Critically evaluate different knowledge representation techniques, highlighting their strengths and weaknesses in different application scenarios.